Question no 3. Complex ion $[Ni(H_2O)_6]^{2+}$

a) Ligands are ions or neutral molecules that donate a lone pair of electrons to a central metal atom/ion to form a coordination complex.
CO is a stronger ligand than H_2O because it is a **π-acceptor ligand**: it donates electron density through σ-bonding and also accepts electron density from the metal's d-orbitals into its empty π* orbitals (back-bonding), which strengthens the metal–ligand bond. H_2O is only a σ-donor.

b) Crystal Field Theory (CFT): In $[Ni(H_2O)_6]^{2+}$, Ni^{2+} (d^8) has an octahedral geometry. The water ligands split the d-orbitals into lower-energy t_{2g} and higher-energy e_g sets. The electronic configuration is $t_{2g}^6 e_g^2$. The transition of an electron from t_{2g} to e_g absorbs light in the visible region (red-orange), so the complementary colour (green) is transmitted, giving the complex its green appearance.

c) Two reasons transition elements readily form complexes:

1. They have small, highly charged ions with vacant d-orbitals to accept electron pairs.
2. They exhibit variable oxidation states, allowing formation of stable coordinate bonds.

4. Silver Extraction and Uses

a) Silver ores are leached with cyanide (e.g., NaCN) because Ag^+ forms a stable soluble complex $[Ag(CN)2]-[Ag(CN)_2]-$, which is extracted from the ore. The reaction:

$$Ag_2S + 4NaCN \rightarrow 2Na[Ag(CN)_2] + Na_2S$$

b) Preparation of $AgNO_3$ from silver:
Dissolve silver in dilute nitric acid:

$$3Ag + 4HNO_3 \rightarrow 3AgNO_3 + NO + 2H_2O$$

Evaporate to obtain crystalline $AgNO_3$.

c) $AgNO_3$ solution stains fingers black due to the reduction of Ag^+ to metallic silver by organic matter (keratin) on the skin. The silver particles deposit, leaving a dark stain that fades slowly.

d) Tollens' reagent preparation:
For preparation of Tollens' reagent dilute NaOH is added to $AgNO_3$ to form Ag_2O, then NH_3 is added until the precipitate dissolves:

$$AgNO_3+NaOH\rightarrow AgOH+NaNO_3$$

$$2AgOH\rightarrow Ag_2O+H_2O$$

$$Ag_2O + 4NH_3 + H_2O \rightarrow 2[Ag(NH_3)_2]^+ + 2OH^-$$

OR – Cement

a) Approximate composition of Portland cement:

- Lime (CaO): 60–67%
- Silica (SiO_2): 17–25%
- Alumina (Al_2O_3): 3–8%
- Iron oxide (Fe_2O_3): 0.5–6%
- Magnesia (MgO), etc.

C_2S, C_3S, C_3A, C_4AF are abbreviations in cement chemistry:

I. C_2S = Dicalcium silicate ($2CaO \cdot SiO_2$)
II. C_3S = Tricalcium silicate ($3CaO \cdot SiO_2$)
III. C_3A = Tricalcium aluminate ($3CaO \cdot Al_2O_3$)
IV. C_4AF = Tetracalcium aluminoferrite ($4CaO \cdot Al_2O_3 \cdot Fe_2O_3$)

b) Six differences between OPC (Ordinary Portland Cement) and PPC (Pozzolana Portland Cement):

Feature	OPC	PPC
Composition	Mainly clinker (95%) + gypsum	Clinker + pozzolanic material (15–30%)
Initial setting time	Faster	Slower
Heat of hydration	Higher	Lower
Durability	Less resistant to chemical attack	More resistant to sulphate and chloride attack
Cost	Lower	Slightly higher
Workability	Moderate	Higher due to finer particles

Acquaintance with Reactions

(a) Nucleophilic substitution in chlorobenzene:
Chlorobenzene is very less reactive towards nucleophilic substitution due to resonance stabilization which shows that chorine in chlorobenzene attached with ring through partial

double bond. Under drastic conditions (e.g., high temperature, pressure, strong base), it undergoes addition-elimination to give phenol or aniline.

(b) Wurtz reaction:
When alkyl halides react with sodium in dry ether to give higher alkanes is known as Wurtz reaction:

$$2R - X + 2Na \xrightarrow{Dry\ ether} R - R + 2NaX$$

(where RR = alkyl group, XX = halogen – usually Cl, Br, or I)

$$2CH_3CH_2Br + 2Na \xrightarrow{Dry\ ether} CH_3CH_2CH_2CH_3 + 2NaBr$$

Limitations:

- Only works for symmetrical alkanes (if two different alkyl halides are used, a mixture of three alkanes is obtained).
- Not suitable for tertiary alkyl halides (they undergo elimination).
- Yields are poor when halides have bulky groups.

Uses:

- Preparation of higher alkanes from lower alkyl halides in the labo

- **(c) Carbylamine test:**
The carbylamine test is a chemical test used to distinguish primary aliphatic and aromatic amines from secondary and tertiary amines.

When a primary amine is heated with chloroform and alcoholic potassium hydroxide (KOH), it forms an isocyanide (carbylamine), which has a very foul, unpleasant odor, it is called carbylamine test.

$$R{-}NH_2 + CHCl_3 + 3KOH \rightarrow R{-}NC + 3KCl + 3H_2O$$

(Primary amine → Alkyl/Aryl isocyanide)

For example,

$$CH_3CH_2NH_2 + CHCl_3 + 3KOH \rightarrow CH_3CH_2NC + 3KCl + 3H_2O$$

Ethyl isocyanide

Positive Result: Foul-smelling isocyanide (carbylamine) – confirms primary amine.

Negative Result: No foul odor – indicates secondary or tertiary amine.

Limitations:

- Only primary amines give a positive test.
- The test is not quantitative.
- Requires careful handling because isocyanides are toxic and foul-smelling.
- Chloroform and alcoholic KOH must be used in proper amounts.

Applications:

- To distinguish primary amines from secondary and tertiary amines.
- Used in qualitative organic analysis for functional group detection.

Sequence Using Five Organic Compounds

Compounds: benzene, benzaldehyde, toluene, cinnamic acid, sodium benzoate.

A possible sequence:

1. **Benzene → Toluene** (Friedel-Crafts alkylation with $CH_3Cl/AlCl_3$).
2. **Toluene → Benzaldehyde** (oxidation with CrO_3 or side-chain chlorination followed by hydrolysis).
3. **Benzaldehyde → Cinnamic acid** (Perkin reaction with acetic anhydride and sodium acetate).
4. **Cinnamic acid → Sodium benzoate** (oxidative decarboxylation)

(Write reaction yourself)

Reaction Sequence for A, B, C, D, E

Given:

$$A \xrightarrow{SOCl_2} B \xrightarrow{NH_3} C \xrightarrow{Br_2/NaOH} D \xrightarrow{NaNO_2/HCl} E$$

E is a primary alcohol with positive iodoform test. Positive iodoform indicates presence of $CH_3CH(OH)$ or CH_3CO- group. Since E is primary alcohol, it must be ethanol (CH_3CH_2OH) because ethanol gives iodoform test. So E = CH_3CH_2OH.

Working backwards:

- D → $NaNO_2$/HCl gives E (primary alcohol). That is a diazotization reaction, so D must be a primary amine (RNH_2) that gives alcohol via diazonium salt. So D = ethylamine ($CH_3CH_2NH_2$).

$$CH_3CH_2NH_2 \xrightarrow{NaNO_2/HCl} CH_3CH_2OH$$

- C → Br_2/NaOH gives D (ethylamine). Br_2/NaOH is Hofmann degradation of amide:

  So C is an amide: propionamide ($CH_3CH_2CONH_2$).

$$CH_3CH_2CONH_2 \xrightarrow{Br_2/NaOH} CH_3CH_2NH_2$$

- B → NH_3 gives C (amide). So B is an acid chloride: propionyl chloride (CH_3CH_2COCl).

$$CH_3CH_2COCl \xrightarrow{NH_3} CH_3CH_2CONH_2$$

- A → $SOCl_2$ gives CH_3CH_2COCl (B). So A is propanoic acid (CH_3CH_2COOH).

  Thus: $CH_3CH_2COOH \xrightarrow{SOCl_2} CH_3CH_2COCl$

- A = CH_3CH_2COOH (propanoic acid)
- B = CH_3CH_2COCl (propionyl chloride)
- C = $CH_3CH_2CONH_2$ (propionamide)
- D = $CH_3CH_2NH_2$ (ethylamine)
- E = CH_3CH_2OH (ethanol)

  Reactions:

1. $CH_3CH_2COOH + SOCl_2 \rightarrow CH_3CH_2COCl + SO_2 + HCl$
2. $CH_3CH_2COCl + 2NH_3 \rightarrow CH_3CH_2CONH_2 + NH_4Cl$
3. $CH_3CH_2CONH_2 + Br_2 + 4NaOH \rightarrow CH_3CH_2NH_2 + 2NaBr + Na_2CO_3 + 2H_2O$
4. $CH_3CH_2NH_2 + NaNO_2 + HCl \rightarrow CH_3CH_2OH + N_2 + NaCl$

Nitrobenzene

a) Nitrobenzene is meta-directing because the $–NO_2$ group is strongly electron-withdrawing by both inductive and resonance effects, deactivating the ortho and para positions more than meta, leading to meta substitution.

(Resonating structures)

b) Reduction in alkaline medium:
In alkaline medium (e.g., Zn/NaOH or Na_2S), nitrobenzene is reduced to azoxybenzene, then azobenzene, etc. For example:

$$C_6H_5NO_2 + 4Zn + 8NaOH \rightarrow C_6H_5N{=}NC_6H_5 + 4Na_2ZnO_2 + 4H_2O$$

(write name also)

c) Two methods to convert nitrobenzene to aniline:

1. **Catalytic hydrogenation:**

$$C_6H_5NO_2 + 3H_2 \xrightarrow{Ni/Pt} C_6H_5NH_2 + 2H_2O$$

2. **Chemical reduction with Sn/HCl or Fe/HCl:**

$$C_6H_5NO_2 + 6[H] \xrightarrow{\text{Sn/HCl}} C_6H_5NH_2 + 2H_2O$$

Group C: Long Answer Questions

Question N. 1

Data:

Expt	[A] (M)	[B] (M)	Initial rate (M/s)
1	0.10	0.10	2×10^{-3}
2	0.20	0.10	4×10^{-3}
3	0.10	0.20	8×10^{-3}

a) Order wrt A: Compare experiment 2 by 1…… . order 1

Order wrt B: Compare expt 1 & 3: calculate → order 2.

b) Overall order = 1 + 2 = 3.

Unit of k: rate = $k[A][B]^2$ → k = rate/($[A][B]^2$) = (M/s)/($M\cdot M^2$) = $M^{-2}\ s^{-1}$. =$mol^{-2}L^2S^{-1}$

c) Rate law: Rate = $k[A][B]^2$.

Using expt 1: $2\times10-3=k(0.10)(0.10)2$

$=k(0.001)2\times10^{-3}$

$=k(0.10)(0.10)^2$

$=k(0.001)$

$k = 2\ M^{-2}\ s^{-1}$.

d) Activation energy (Ea): Minimum additional energy required for reactants to form an activated complex is known as activation energy.

$$E_a = E_T - E_R$$

Energy profile diagram: (sketch) Reactants → transition state (higher energy) → products. Catalyst lowers the activation energy by providing an alternate pathway.

OR – Solubility Product

a) For $Ag_2CrO_4 \rightleftharpoons 2Ag^+ + CrO_4^{2-}$,

solubility $s = 6.5\times10^{-5}$ M.

$K_{sp} = [Ag^+]^2[CrO_4^{2-}]$

$= (2s)^2 \times (s)$

$= 4s^3$

$= 4(6.5\times10^{-5})^3$

$= 4\times274.625\times10^{-15}$

$= 1.0985\times10^{-12} \approx 1.1\times10^{-12}$

b) Mixing 100 mL of 0.01 M $AgNO_3$ + 100 mL of 0.002 M K_2CrO_4:

Total volume = 200 mL.

$[Ag^+]$ after mixing = (0.01)/2 = 0.005 M; $[CrO_4^{2-}]$ = (0.002)/2 = 0.001 M.

Ionic product = $(0.005)^2(0.001) = 2.5\times10^{-8}$.

Since 2.5×10^{-8} > Ksp (1.1×10^{-12}), precipitate forms.

c) In 0.1 M $AgNO_3$, solubility s of Ag_2CrO_4:

$[Ag^+] = 0.1 + 2s \approx 0.1$ (due to common ion), $[CrO_4^{2-}] = s$.

$Ksp = (0.1)^2 \cdot s = 0.01\ s$

$\Rightarrow s = Ksp / 0.01$

$= (1.1\times10^{-12})/0.01 = 1.1\times10^{-10}$ M.

Common ion effect: solubility decreases dramatically in presence of Ag^+.

d) Applications in qualitative analysis:

1. Selective precipitation of ions (e.g., Group I chlorides).
2. Separation of cations by controlling ion concentration (e.g., H_2S precipitation in acidic/alkaline medium).

Conversions

(a) 2-Nitropropane to 1-nitropropane:

2-nitropropane → isopropylamine → isopropyl alcohol (via diazonium) → propene (dehydration) → 1-bromopropane (HBr/peroxide) → 1-nitropropane

OR

- 2-Nitropropane → (acid hydrolysis) → acetone → (reduction) → 2-propanol → (dehydration) → propene → (HBr/peroxide) → 1-bromopropane → ($NaNO_2$) → 1-nitropropane.

(Write only in reaction form)

- **(b) Ethanol to Propanoic acid:**

  Ethanol → ethanal (oxidation) → (cyanohydrin formation with HCN) → 2-hydroxypropanenitrile → hydrolysis → 2-hydroxypropanoic acid → (dehydration) → propanoic acid. OR

- Ethanol → ethanal → (cyanohydrin) → 2-hydroxypropanenitrile → hydrolysis → 2-hydroxypropanoic acid → (reduction with HI) → propanoic acid.

OR

- Ethanol → ethyl chloride ($SOCl_2$) → (Mg) → ethylmagnesium chloride → (CO_2) → propanoic acid.

  (Write any one type reaction only)

  (c) Nitrobenzene to benzene:
  Nitrobenzene → (reduction) → aniline → (diazotization with $NaNO_2$/HCl) → benzene diazonium chloride → (reduction with H_3PO_2 or ethanol) → benzene.

  (Write reaction only)

  (d) Acetone to acetaldehyde:

  Acetone → (reduction with $NaBH_4$) → isopropanol → (dehydration with H_2SO_4) → propene → (ozonolysis) → acetaldehyde + formaldehyde. Then separate.

  (Write reaction only)

  (Other methods can also be used.)

Compounds A and B

Assuming A = acetaldehyde (CH_3CHO) and B = acetone (CH_3COCH_3).

a) Aldol condensation: Both can undergo aldol condensation.

- Acetaldehyde (A) gives 3-hydroxybutanal (aldol).

$$CH_3CHO \xrightarrow{OH^-} CH_3CH(OH)CH_2CHO$$

Acetone (B) gives diacetone alcohol (4-hydroxy-4-methylpentan-2-one).
Example:

$$CH_3COCH_3 \xrightarrow{OH^-} CH_3CH(OH)CH_2COCH_3$$

b) Iodoform test: Both give positive iodoform test because they have CH_3CO- group (acetone) or $CH_3CH(OH)$- after oxidation (acetaldehyde).

Reaction:

$$CH_3COCH_3 + 3I_2 + 4NaOH \rightarrow CHI_3 + CH_3COONa + 3NaI + 3H_2O$$

c) Compound A (acetaldehyde) with HCN:

$$CH_3CHO + HCN \rightarrow CH_3CH(OH)CN \qquad \text{(C)}$$

Complete acid hydrolysis:

$$CH3CH(OH)CN + 2H2O + H^+ \rightarrow CH3CH(OH)COOH + NH4^+ \qquad \text{(D)}$$

Thus C = lactonitrile (2-hydroxypropanenitrile), D = α-hydroxypropanoic acid (lactic acid).